Logos Experiment 13 Report

Medium Scaling with Weight Tying + SwiGLU

1. Experiment Purpose

Experiment 13 starts the scaling validation phase of the Logos project. Earlier experiments established that Weight Tying + SwiGLU improves compact model quality, while the wider-shallower recipe improves speed and energy efficiency. This experiment tests whether the efficient Logos recipe still works when the model is scaled to a medium size with more data and a longer context length.

The core research question is: Does the best efficient design discovered in small-scale ablations continue to improve language-modeling quality when scaled to a larger model and a larger OpenWebText subset?

2. Experimental Setup

Component	Value
Run name	logos_exp13_medium_scaling_weight_tying_swiglu
Model family	Decoder-only causal language model
Dataset	OpenWebText subset
Dataset size	50,000 samples
Tokenizer	GPT-2 tokenizer
Objective	Next-token prediction
Architecture recipe	Weight Tying + SwiGLU
Training iterations	3000

3. Model Configuration

Parameter	Value
Total parameters	51,202,896
Layers	8
Embedding dimension	512
Attention heads	8
Head dimension	64
Context length	512 tokens
Batch size	32
Tokens per step	16,384
Feed-forward network	SwiGLU
Weight tying	Enabled
Normalization	LayerNorm

4. Final Results

Metric	Experiment 13 Result
Best iteration	3000

Metric	Experiment 13 Result
Best validation loss	5.0266
Final validation loss	5.0421
Final validation perplexity	154.79
Checkpoint size	195.36 MB
Peak VRAM	13.09 GB
Training time	380.79 sec / 6.35 min
Generation speed	75.82 tokens/sec
Generation latency	13.19 ms/token
Average GPU power	282.32 W
Estimated GPU energy	29.70 Wh

5. Comparison with Previous Best Models

Metric	Exp 11 Best Quality	Exp 12 Best Efficiency	Exp 13 Medium Scaling
Parameters	19.24M	18.54M	51.20M
Dataset samples	10k	10k	50k
Context length	256	256	512
Best val loss	5.2932	5.3469	5.0266
Final val loss	5.3123	5.3470	5.0421
Final perplexity	202.82	210.00	154.79
Checkpoint size	73.45 MB	70.76 MB	195.36 MB
Peak VRAM	11.18 GB	10.54 GB	13.09 GB

Experiment 13 substantially improves validation quality. Perplexity decreased from 202.82 in Experiment 11 and 210.00 in Experiment 12 to 154.79 in the medium-scaled model. This confirms that the Weight Tying + SwiGLU recipe remains effective when scaled to a larger parameter count and larger data subset.

6. Scientific Interpretation

The main result is that scaling the efficient architecture improves language-modeling capability significantly. The parameter count increased from around 19M to 51M, and the dataset increased from 10k to 50k samples. In return, the model achieved a much lower validation loss and perplexity while using only about 13.09 GB VRAM on a 48 GB GPU system.

This suggests that the Logos efficiency recipe is not limited to very small models. The medium model retained the same core design principles - weight tying for parameter efficiency and SwiGLU for stronger feed-forward representation - while showing clear capability gains from scale.

7. Tradeoff Analysis

Positive Outcome	Tradeoff / Cost
Lowest perplexity so far: 154.79	Checkpoint grew to 195.36 MB
Best validation loss so far: 5.0266	Training energy increased to 29.70 Wh
Only 13.09 GB VRAM used despite 51.20M params	Generation latency remained 13.19 ms/token
Strong scaling result on 50k samples	Training time increased to 6.35 minutes

8. Conclusion

Experiment 13 is a major success for the scaling phase of Logos. It proves that the efficient recipe found in earlier ablations - especially Weight Tying + SwiGLU - can scale to a medium-sized model and significantly improve validation quality. The model achieved the best perplexity so far while still using far less than the available 48 GB VRAM.

The next logical step is to test a larger 100M+ parameter model, using the same measurement framework, to see whether the capability-resource trend continues at a larger scale.

9. Recommended Next Step

Run Experiment 14 as a larger scaling test using N_EMBD = 768, N_LAYER = 12, N_HEAD = 12, BLOCK_SIZE = 512, BATCH_SIZE = 16, Weight Tying enabled, SwiGLU enabled, and 50k to 100k OpenWebText samples. This will test whether the Logos scaling trend continues toward the 100M to 150M parameter range.